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Research and Application of Ultra-High Pressure Intelligent Well Control Technology for Ultra-Deep Carbonate Rocks

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Abstract

The exploration and development of the Tarim Oilfield is vigorously advancing into ultra-deep layers. Since 2021, more than 200 deep wells of the 8000m class have been completed. Ultra-deep drilling has brought about risk challenges such as ultra-high pressure, ultra-high temperature, and high hydrogen sulfide content. At present, the maximum formation pressure tested in the Tarim Oilfield is 182 MPa, and the maximum hydrogen sulfide content in the formation reaches 41×10^4 ppm. Therefore, the ultra-high pressure intelligent well control disposal technology for ultra-deep carbonate rocks and supporting equipment have been developed.

Tarim Oilfield has taken various technical measures to solve the risk of ultra-high pressure well control. Firstly, in terms of engineering design, Tarim Oilfield has formed an integrated risk intelligent identification technology for "five specials" (special surface, special lithology, special geological phenomenon, special fluid, special temperature and pressure field), and adopted a comprehensive prediction technology for "five pressures" (pore pressure, collapse pressure, loss pressure, closure pressure and fracture pressure) to predict formation pressure intelligently. Secondly, in terms of well control technology, the application of controlled pressure drilling, DROC and other technologies has successfully ensured the safe drilling of ultra-high pressure Wells in this block. Finally, supporting equipment for ultra-high pressure intelligent well control has been developed, and a remote management and control support center for oilfield drilling and completion has been established, enabling capabilities such as real-time monitoring, abnormal alarm, emergency response, and remote command, thus achieving intrinsic safety in well control.

Well control disposal technology and ultra-high pressure intelligent well control supporting equipment technology have been successfully applied in 51 ultra-deep Wells in the Tarim Oilfield. A series of equipment has been developed, including 175MPa ultra-high pressure blowout preventer stacks, 175MPa casing heads, 175MPa electric choke/kill manifolds and control systems, and 175MPa Christmas tree. In addition, a risk assessment technology for an ultra-high pressure well control equipment, a full life cycle management system for well control equipment, a one-click automatic alarm for overflow, a one-click shear well shutdown, and a one-click well shutdown pressure relief and blowout ignition system have been established. It has successfully handled complex accidents of high casing pressure and overflow in 17 wells, with the maximum shut-in pressure reaching 47.57MPa and the maximum overflow disposal pressure reaching 64.4MPa. These efforts have ensured well control safety in the block and achieved high oil production.

The promotion and application of the ultra-deep and ultra-high pressure intelligent well control risk disposal technology and supporting equipment in the Tarim Oilfield have provided excellent engineering cases and experiences for the development of ultra-high pressure intelligent well control. This paper also discusses the development trend of intelligent well control treatment technology under extreme working conditions.

Key words: ultra-deep layer, carbonate rock, ultra-high pressure, well control disposal, well control equipment

Introduction

The Tarim Basin is an important base for deep and ultra-deep oil and gas exploration and development in China. The ultra-deep carbonate strata in the Tarim Basin have undergone eight phases of orogenic movements, resulting in extremely complex geological and engineering conditions. They generally feature high temperature, high pressure, high sulfur content, and ultra-depth, making exploration and development extremely challenging. At present, the highest formation pressure tested in the Tarim Oilfield is 182MPa, and the highest formation hydrogen sulfide sulfur content is 41×10^4 ppm. Since 2015, the Tarim Oilfield has

encountered abnormal high pressure 14 times during drilling (well closure pressure >20MPa), and the well control risk is very high. The average burial depth of the target layer in the Tarim Basin exceeds 6000m. Since 2021, drilling has gradually reached a depth of over 8000m. Currently, more than 200 Wells with a depth of over 8000m have been completed, among which the depth of the SSDK1 Well reaches 10910m (Yan and Li 2008, Wu et al.2022, Wu et al.2020).

Drilling in ultra-deep carbonate rock formations is confronted with inaccurate formation pressure prediction, high overflow shut-off pressure caused by encountering abnormally high-pressure formations, and significant well control safety risks (Song et al.2022, Li et al.2019, Duan et al. 2019). The fluid medium is diverse and passes through multiple mandatory sealing points, and the number of layers of the wellbore structure is limited. Drilling fluid overflow and leakage are frequently converted and difficult to handle. Especially in carbonate fracture-cave reservoirs, it is highly prone to fluid loss due to backflow, with a fast oil and gas displacement rate and high well control risks (Ma 2012, Li et al. 2008, Xiong et al. 2010, Shi et al.2006, Yu et al. 2014, Lu et al. 2017, Mao 2020, Xu et al. 2016). At the same time, high-sulfur, high-speed and high-density fluids cause severe damage to drilling tools and well control equipment (Wu et al 2010, Li et al.2020, Fan et al. 2020, He et al. 2019, Zhang et al. 2019, Zheng et al. 2019).

There are several difficulties in the construction of ultra-deep drilling, such as casing head sinking and yield failure under high-load conditions, sealing failure of ultra-high temperature well control devices, immature matching of 175MPa oil and gas production wellheads, insufficient strength and durability of internal blowout prevention tools, and the lack of risk assessment methods for ultra-deep well control equipment systems (Teng et al 2018, Wang et al. 2020, Zhang et al. 2018, Zhang et al. 2021, Li et al.2013, Li et al. 2008, Zhang et al. 2012, Gong et al. 2012).

Ultra-High Pressure Risk Well Control Technical Measures

Well Control Risk and Pressure Prediction Technology

"Five Specials" Risk Identification Technology. In order to systematically and scientifically analyze the impact of special factors in ultra-deep layers on drilling engineering, and to enhance the comprehensiveness of understanding oil and gas reservoirs and the assessment of well control safety risks, Tarim Oilfield has developed the "Five Specials" risk identification technology, which refers to special surface, special lithologic bodies, special geological phenomena, special fluids, and special temperature and pressure fields (Fig.1).

Well control risks are identified through special surface. It means that the well site layout direction and road setting may cause potential harm to high-risk places such as forest belts, residential areas, and wetland protection areas. Well control risks are identified through special lithological bodies. It means that the shrinkage of gypsum salt rock and under-compacted soft mudstone causes excessive drilling pumping. Carbonate salt venting and leakage may cause the drilling fluid column pressure to be unable to balance the formation pressure. Well control risks are identified through special geological traces. When drilling through faults, cracks, unconformities and other geological structures, it is necessary to optimize the wellbore trajectory to avoid low leakage and fracture pressure formations, and prevent and control the risk of leakage and overflow caused by drilling fluid loss. Well control risks are identified through special fluids. When drilling may encounter hydrogen sulfide, carbon dioxide and shallow non-target oil and gas, it is necessary to use sulfur-resistant drilling and completion tools, select a reasonable drilling fluid pH value, and design a reasonable wellbore structure. Well control risks are identified through special temperature and pressure fields. When drilling may encounter abnormally high or low temperature and pressure fields, it is necessary to design a reasonable drilling fluid density and wellbore structure, and use matching casing, wellhead equipment, and internal spraying tools to prevent well control. Through the "Five Specials" technology,

well control risks are comprehensively and systematically identified, and preventive measures are taken in advance.

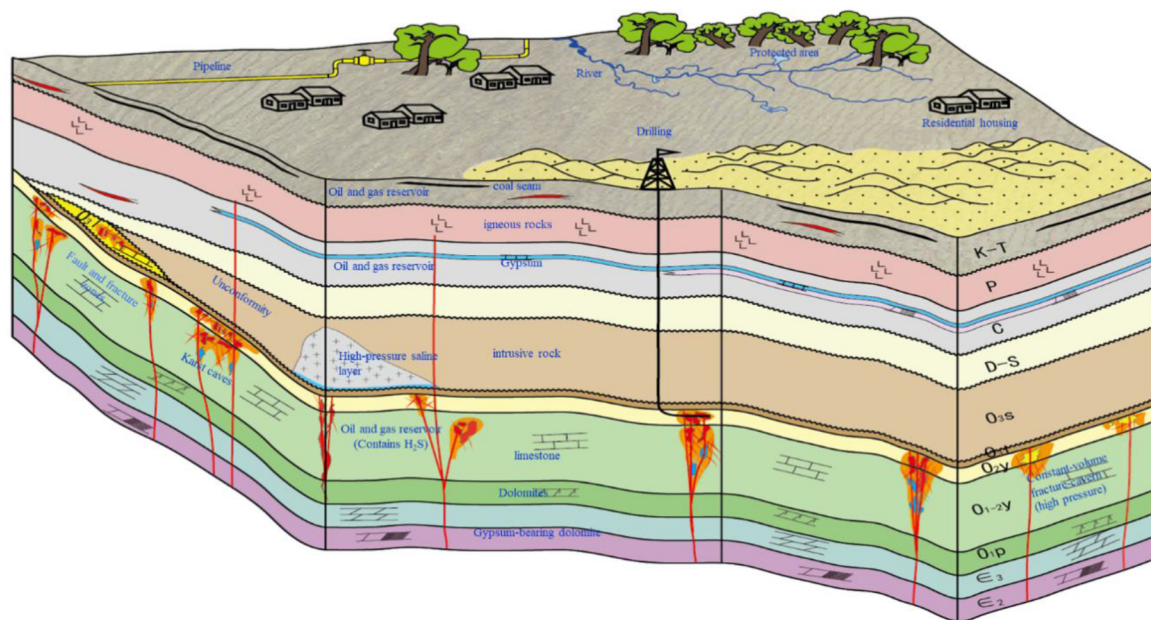


Figure 1—"Five Specials" risk identification in the Tabei area of Tarim Basin

Five pressure coefficient prediction techniques. The traditional prediction of the three pressure coefficients (pore pressure, collapse pressure, fracture pressure) before drilling lacks detailed data on the loss pressure of the entire wellbore, resulting in an unclear recognition of the causes and distribution patterns of abnormal high pressure and making it difficult to avoid complex well control situations. Based on the prediction of three pressure coefficients, the prediction of loss pressure and closure pressure is added to establish an upgraded five-pressure coefficient prediction technology. That is the prediction of pore pressure, collapse pressure, loss pressure, closure pressure and fracture pressure. Based on seismic data, VSP and the measured trend control of adjacent wells are adopted to improve the prediction accuracy of pore pressure. The relationship between loss pressure difference and loss rate is established by fine division and stratification, and the leakage pressure prediction method is determined. Drilling parameters and drilling fluid performance are optimized in a timely manner to avoid excessive bottom-hole pressure difference and surge pressure, which may lead to lost circulation and well control risks.

Manage pressure drilling technology (MPD)

MPD ensures that the bottom hole pressure is controlled within the designed range through the reasonable adjustment of the throttle valve in the surface manifold, and can precisely control the annular pressure profile of the entire wellbore. The dynamic changes of bottom-hole pressure in real-time during the drilling process is monitored, effectively prevents well leakage and well gushing in the easily leaking strata underground. This technology solves the problem of frequent switching between leakage and overflow caused by the narrow pressure window in carbonate rock formations, and achieves effective protection of oil and gas layers.

Due to the influence of geological structure, development degree, reservoir physical properties, gas-oil ratio, and hydrogen sulfide, some Wells of carbonate rocks frequently experience complex downhole conditions such as large well loss and high casing pressure during the drilling process of the target layer, which makes it impossible to fully achieve the geological goals. By combining the dual-density unconventional drilling and conventional MPD technology, the technical bottlenecks that it is difficult

to achieve geological targets due to depletion and cross-flow has been broken through. In 2023, 9 wells operations were implemented, maximizing the exposure of reservoirs and the achievement of geological objectives. In 2023, 61 wells reached the designed depth, accounting for 72.6% of the completed carbonate rock wells, an increase of 50.8% compared with 2022.

Ultra-Deep Wellhead Bearing Enhancement Technology

The bearing capacity of the casing head steps is enhanced, and the design of the inner cavity of the casing head is optimized. Invalid steps are eliminated, the width of the seating and hanging steps of the pressure test plug is expanded, the bearing area is increased, and the compressive strength of the bearing steps during the pressure test is reduced. A groove with a load-reducing function is designed at the position of the main body bearing step, and a 718 nickel-based alloy ring is inlaid in the groove, which increases the yield strength by 1.6 times (Fig.2) and avoids yield deformation of the bearing contact surface.

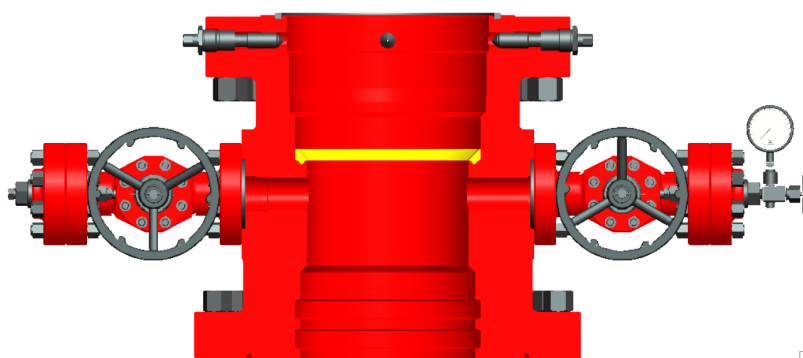


Figure 2—Schematic diagram of improving the yield strength of the contact surface

Multi-point support solves the dilemma of the surface casing bearing force alone, and an innovative method for stabilizing and supporting the surface casing head is proposed. The bearing capacity of the cement base surface of the cellar is improved, and the strength of the bottom tray and support blocks of the surface casing head is reinforced. The installation cutting height of the surface casing is precisely calculated. A combined cushion block is designed to realize the joint bearing of the cement base surface of the cellar and the surface casing, so as to avoid the sinking of the surface casing and the uneven settlement of the wellhead (Fig.3).

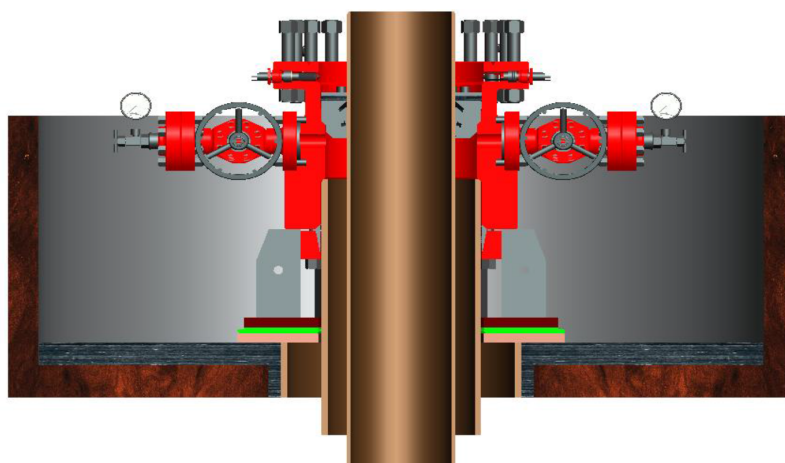


Figure 3—Schematic diagram of the force on the surface casing

Supporting Equipment for Ultra-High Pressure Well Control

A series of equipment has been developed, including 175MPa ultra-high pressure blowout preventer stacks, 175MPa electric oil nozzle, 175MPa HH-class Y-type Christmas tree, 175MPa casing heads, 175MPa choke and kill manifolds, as well as supporting high-performance internal blowout prevention tools and new efficient large-flow liquid-gas separators.

Development of 175MPa ultra-high pressure wellhead equipment. A 175MPa ultra-high-pressure blowout preventer assembly was successfully developed. By optimizing the rubber formula and incorporating a multi-stage low-temperature mixing process, a new rubber core skeleton structure with side wing protection and front wing wrapping was designed. A ram blowout preventer sealant with a pressure resistance of 175MPa and an extreme high temperature of 177°C was developed. By optimizing the forging process and combining with rapid intelligent quenching, the problems of looseness, segregation and microstructural uniformity of large components have been solved, ensuring the stable performance of the pressure-bearing base material of the blowout preventer. The structural design scheme of the 35-105 annular blowout preventer has been optimized. Through hyper elastic dynamic simulation analysis, with comprehensive consideration of sealing specific pressure and stress-strain, the geometric configuration of the packing element, the metal skeleton, and the performance parameters of the hyperelastic body have been adjusted. The 35-105 annular blowout preventer was successfully designed and manufactured (Fig.4).



Figure 4—175MPa Blowout Preventer Stack

Based on the micro-cutting theory, the erosion law of the two-phase component was revealed, and a 175MPa electric oil nozzle was developed. By optimizing the size of hard particles and the sintering process, the bonding performance at the two-phase interface was enhanced, and the erosion resistance of the nozzle material was increased by 40%. An improved anti-erosion material of X was formed, solving the technical problem of erosion damage to the valve core of the throttle valve under a high-speed flow environment of 175MPa. An electric nozzle with a low switching torque under a throttling pressure difference of 175MPa was developed. The switching torque is 20% lower than that of similar products, meeting the demand for precise production adjustment over a wide range of 100000 to 4 million cubic meters per day. It also features remote production adjustment function (Fig.5).



Figure 5—Ultra-high pressure electric oil nozzle

A 175MPa HH-class Y-type Christmas tree has been developed. Using the nonlinear contact theory, combined with thermal-mechanical coupling and elastic-plastic constitutive models, multi-field and multi-scale analyses were carried out to reveal the failure mechanism of key components and sealing parts of the Christmas Tree under the conditions of 175MPa and 180°C. Through the optimization of micro-alloys, corrosion-resistant elements, and heat treatment processes, the problem of matching high strength and high sulfur resistance for thick-walled parts was overcome, and an 85K improved alloy steel material was innovatively developed. Thus, the L-X (-46~+180°C), HH-class Y-type Christmas tree was successfully developed (Fig.6).



Figure 6—175MPa Christmas Tree

A 175MPa casing head has been developed. A failure assessment model considering the coupling of multiple factors such as heat, force, shock and vibration has been constructed. Based on intelligent algorithms and experimental evaluation, a reliability analysis and optimization method for key components has been established. Combined with large-size, high-wall-thickness alloy forging and heat treatment technology, a high-load-bearing casing head with a hanging load of 860t and an ultimate composite load-bearing capacity of 1600t has been successfully manufactured (Fig.7).



Figure 7—The world's first set of 175MPa casing heads

A 175MPa choke and kill manifold has been developed. By optimizing the designs of pressure self-tightening floating seal, high-performance transmission mechanism, and valve cavity volume balance,

the stiffness of components is improved, the contact area is reduced, and the surface treatment process is optimized. A manual slab valve with low torque and high load-bearing capacity has been developed, achieving the technical targets of 18 turns for opening and closing and a torque of 250N·m. A high-linear throttle valve and a high-reliability check valve suitable for high-pressure, high-speed, and high-temperature flow conditions have also been developed. Based on the simulation analysis of manifold impact and vibration under gas-liquid-solid three-phase flow conditions, the flow field of the manifold is optimized, and a five-channel two-stage choke manifold and a kill manifold with flow channels that have a throttling function have been successfully developed (Fig.8).

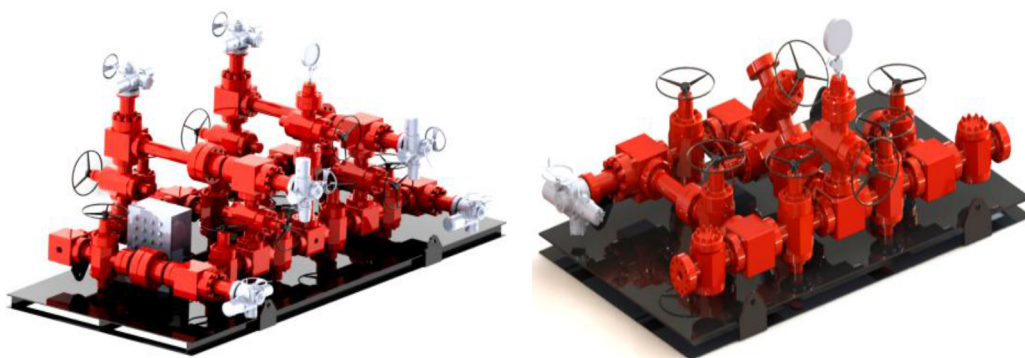


Figure 8—175MPa Kill Manifold(Left)and 175MPa Choke Manifold(Right)

High-performance internal blowout prevention tool. A high-load 6-5/8" cock has been developed. High-strength low-alloy steel is selected as the main body material. The knob size is optimized, the knob strength is enhanced, and a double sealing ring is designed to improve the sealing reliability of the knob. Finite element analysis is used to optimize the fillet size of the right-angle step of the cock, reducing the internal stress concentration of the cock. The sealing surface is sprayed with tungsten carbide alloy to improve wear resistance and corrosion resistance (Fig.9). The tensile strength of the new high-load cock is 15% higher than that of ordinary cocks, and the impact energy is increased by 42%.

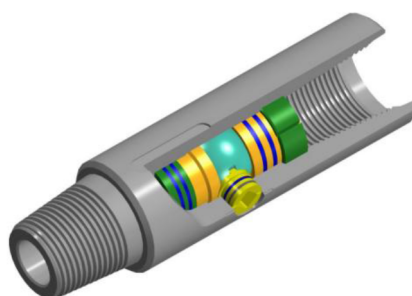


Figure 9—Diagram of the improved cock

New-type High-efficiency Large-capacity Liquid-gas Separator. Conventional liquid-gas separator equipment is confronted with practical problems such as insufficient processing capacity, severe vibration leading to easy failure of pipeline connections, easy erosion due to pipeline bending, high risk of leakage, and rust and corrosion inside the separator and pipelines, which affect the separation efficiency. By increasing the tank diameter of the separator to 1.8m and the pipeline size to 350mm, a first-stage separation partition was designed after finite element simulation analysis, which blocked the erosion of the inner wall of the separator tank and improved the shaking degree of the separator. The spacing of the baffle plates (550mm) and the horizontal angle (16°) were optimized to improve the separation efficiency. Using the blind-end water head to increase resistance, an anti-erosion structure was innovatively designed to reduce the pipeline

erosion rate. A liquid-gas separator with a processing capacity of $100 \times 10^4 \text{ m}^3/\text{d}$ was successfully developed (Fig.10).



Figure 10—High-capacity liquid-gas separator

Risk Assessment Technology for Ultra-Deep Well Control Equipment

The harsh working conditions of ultra-high temperature, ultra-high pressure, high sulfur content, and designed oil-based mud pose challenges to the metal and non-metallic materials of well control equipment. Metal components need to meet the strength standard of 75K steel grade, but high-grade steel materials show significantly increased susceptibility to sulfide stress corrosion cracking (SSCC) in H_2S environments. Rubber seals accelerate H_2S penetration due to volume swelling under high temperature and pressure, causing internal cracks and degradation of mechanical properties. To address this, systematic research has been conducted on the applicability of metal materials and rubber seals under extreme working conditions. By simulating the H_2S environment at the wellhead, high-temperature and high-pressure aging tests on ram packing elements and BT sealing rings have been carried out. Based on numerical simulations and testing, the R&D team focused on issues such as the ultimate load-bearing capacity, sealing reliability, and vibration of well control equipment. Reliability analyses of six major components were conducted. These components include the casing head, blowout preventer, internal blowout preventer tool, choke valve, gas-liquid separator, and sealing flange. The evaluation results indicate that the materials and designs of the current well control equipment meet safety requirements under extreme operating conditions. The evaluation results demonstrated that the materials and designs of current well control equipment meet safety requirements under extreme operating conditions. The assessment results indicate that the current materials and design of the well control equipment meet the safety requirements under extreme working conditions of this well. In response to the risk of pressure test leakage in on-site large-size wellhead flange connections, a magnetic-acoustic flange seal detection system was developed and tested. By detecting the stress of bolts, the stress changes of the main sealing surface of the steel ring were inferred. Field tests showed that this device could monitor the preload of bolts in a timely manner and give early warnings of the risk of flange connection seal failure.

Digital and Intelligent Well Control Technology

A well control online platform has been developed, realizing intelligent management such as registration, training, electronic teaching materials, special training, examinations, big data analysis, and certificate sharing. Combined with VR simulation equipment, it can realize simulated well shutdown drills under various working conditions, comprehensively improving the capabilities of well control technicians (Fig.11).



Figure 11—Well control operation simulator

The business chain for well control equipment management has been fully connected, enabling intelligent well matching and automatic tracking of accessories. Based on well control design, equipment demand, workshop inventory and other factors, a well matching list can be generated with one click, which improves the accuracy and efficiency of equipment well matching by 9 times. The entire process of plan approval, order signing, and material arrival status is tracked, realizing automatic summary and automatic inventory balancing of accessories. The submission of material demand plans can be completed within 2 hours, with efficiency increased by 12 times. A remote management and control support center for oilfield drilling and completion operations has been established to consolidate the intelligent management and control platform. A remote management and control support system featuring "one center, two platforms, and three major scenarios" has been built, equipped with capabilities such as real-time monitoring, abnormal alarm, emergency response, and remote command (Fig.12).

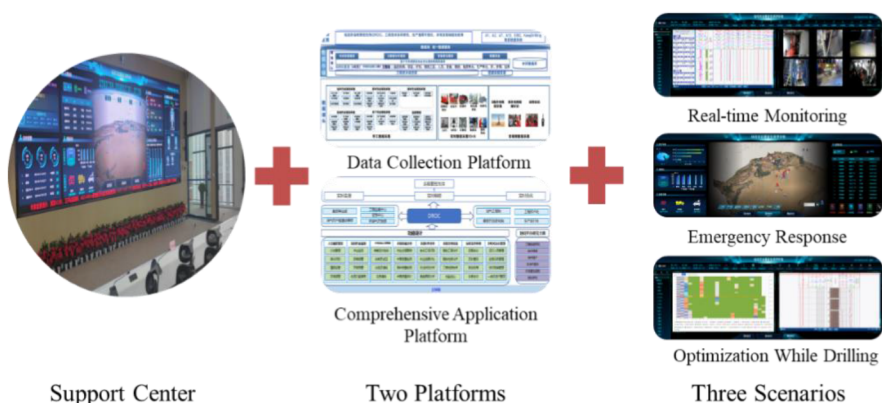


Figure 12—Drilling & Completion Remote Operation Center(DROC)

An intelligent alarm model for overflow and leakage risks under multiple working conditions is developed to strengthen the risk prevention and control barrier. Based on real-time data, using the large model and expert experience approach, alarm models for well leakage and overflow under working conditions such as drilling and tripping are developed, with an overall accuracy rate of 80%. At the same time, the one-click alarm signal from on-site mud logging is integrated to improve the overflow alarm function. A supporting on-site early warning working mechanism is established to realize early warning upon the appearance of signs of engineering abnormalities and intervention at the first sign, effectively enhancing risk prevention and control capabilities.

The integrated drilling and logging liquid level monitoring system is applied to monitor the liquid level status underground in real time. The integrated drilling and logging liquid level monitoring technology is comprehensively promoted to realize remote-controlled automatic measurement and accurate measurement under the well-opening condition. Connecting with the comprehensive mud logging instrument system, and

the data is transmitted to the DROC system for real-time monitoring and early warning of the downhole liquid level. A supporting electronic liquid level monitoring system is equipped to improve alarm accuracy. A pull-rope displacement sensor is configured to monitor the displacement change of the scale through electrical signals. A composite filtering algorithm is introduced to eliminate the error caused by the agitator's bypass flow, enabling real-time and accurate monitoring of liquid level data. The system features automatic sound and light alarm functions as well as intelligent monitoring capabilities (Fig.13 and Fig.14).



Figure 13—Audio-visual alarm system



Figure 14—Intelligent Drilling Liquid Level Monitor

A real-time monitoring system for the switch status and position of well control equipment is developed to realize remote control of the equipment. It enables real-time monitoring, remote transmission, and automatic alarm of information such as the switch status of wellhead equipment and choke manifold, the oil tank level of the control system, system pressure, and the position of three-position four-way directional control valve. In case of emergency, "one-click" remote control operation can be performed within the range of 300-1000m, achieving a 100% success rate of correct well shut-in (Fig.15). An emergency shear one-click well shut-in control system is developed to avoid misoperation in emergency situations. To ensure the safe and reliable operation of the shear ram blowout preventer in emergencies, the oilfield has developed an emergency shear one-click well shut-in control system. Using the principle of progressive control, it changes the five-step shearing operation into one-click linkage; The on-site test completed one-click shearing within 35 seconds, ensuring the accuracy of personnel operation in emergency situations (Fig.16). A remote one-click well shut-in, pressure relief, blowout release and ignition system under extreme working conditions is developed to ensure personnel safety. To ensure the safety of near-wellhead equipment and personnel under extreme working conditions, a one-click well shut-in, pressure relief, blowout release and ignition system has been developed. One-click orderly start of the blowout valve opening, automatic ignition, and closing of the annular preventer, which effectively protects wellhead equipment and enables the rapid evacuation of personnel from the drilling platform. It is equipped with electric control structures at four different positions,

featuring a 500-meter remote control function and a graphical operation mode, with fast switching actions (Fig.17).

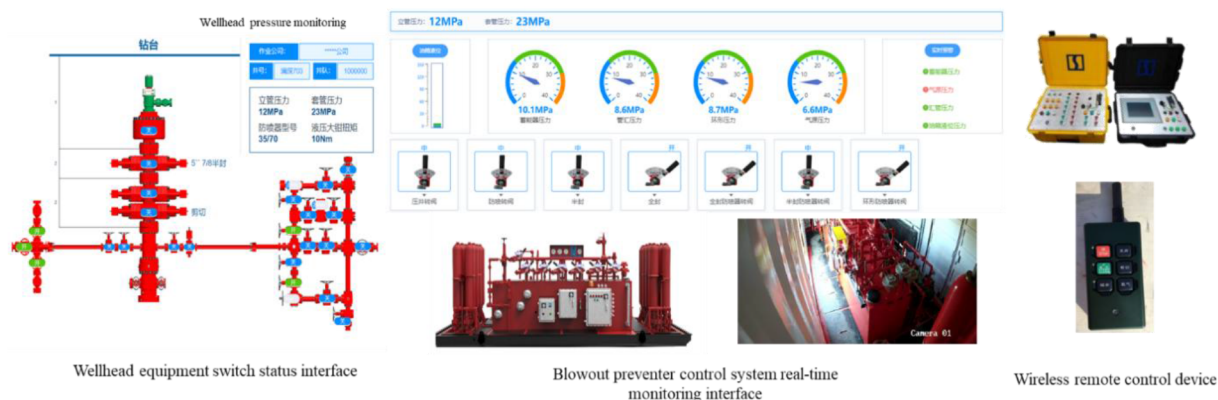


Figure 15—Real-time monitoring and control system for the switch status and position of well control equipment



Figure 16—Indoor Test of Emergency Shear One-Click Well Shut-In Control System



Figure 17—Remote one-button shut-in, pressure relief, and blowout ignition system under extreme working conditions

Multi-parameter and multi-scenario intelligent display assists in efficient remote decision-making and command. By encapsulating core algorithms and deploying on the DROC platform in the cloud, integrating drilling process parameters, and simulating and optimizing well killing techniques, real-time linkage and dynamic tracking from simulation calculation to on-site construction are realized, providing references for well control experts in real-time command.

A high-performance, automated, and electrified well control equipment is developed to improve the efficiency of complex disposal. Based on the 175MPa project, an electrically controlled choke manifold has been developed, accompanied by an intelligent operation control system and a remote-control room.

A choke valve core with erosion resistance and high linearity suitable for well killing conditions has been developed, and an electric (hydraulic) servo drive system with high positioning accuracy, fast response speed, and strong anti-interference capability has been configured, which can intelligently identify blockages and automatically clear them (Fig.18).



(a)175MPa Electric Choke Manifold and Control System (b)175MPa Remote Intelligent Control System (c)Full Bore New-Type Choke Valve

Figure 18—High-performance automated electric well control equipment

The well control technology for ultra-deep carbonate rocks have successfully handled complex accidents of high casing pressure and overflow in 17 wells, with the maximum shut-in pressure reaching 47.57MPa and the maximum overflow disposal pressure reaching 64.4MPa. These efforts have ensured well control safety in the block and achieved high oil production (Table1).

Table 1—Overflow statistics of Tarim Oilfield in recent years

No.	Well Number	Well Type	Design Well Depth (m)	Well Depth at Overflow (m)	Operating Conditions at Overflow	Design Density of Overflow Layer(g/cm ³)	Treatment Density of Overflow Layer(g/cm ³)	Overflow Volume (m ³)	Overflow Fluid Properties	Main Treatment Method	Treatment Time (h)	Direct Cause of Overflow	Indirect Cause of Overflow
1	FD1	Directional well	8697	8190.00	Drilling	1.22-1.95	2.04	0.5	Oil and Gas	Annular pressure back method	16.75	High-pressure carbonate reservoir with low mud density	None
2	MS703	Directional well	8635	8146.01	Drilling	1.10-1.35	1.8	0.8	Natural gas	Annular pressure back method	22.75	Abnormally high-pressure oil, gas, and water reservoir	Inaccurate geological prediction
3	ZG172-H2	Horizontal well	6767	6767	Tripping the drill pipe	1.10-1.40	1.7	0.1	Drilling fluid	Engineer's method, Managed Pressure Drilling	5.32	Failure of downhole tool isolation valves leading to oil and gas intrusion into the wellbore and reduced wellbore mud density	The upper wellbore mud column pressure cannot balance the formation pressure
4	KL801-H	Horizontal well	5066	2245.00	Drilling	1.98-2.20	2.2	1.20	Natural gas	Engineer's method	17.17	Low drilling fluid density leading to gas intrusion	Circulation density reduction
5	GL5	Directional well	8210	7955	Drilling	1.50-2.20	1.9	2.9	Oil and Gas	Annular pressure back method, Managed Pressure Drilling	9	Abnormally high-pressure carbonate reservoir	Low mud density
6	FD2	Directional well	9180	8570.03	Drilling	1.8-2.15	1.8	0.5	Natural gas	Annular pressure back method, Managed Pressure Drilling	4	Abnormally high-pressure oil, gas, and water reservoir	None
7	KL801-H井	Horizontal well	5066	3031.5	Reaming	1.9-2.10	2.04	0.7	Natural gas	Driller's method, Managed Pressure Drilling	13	Low drilling fluid density	Overflow and leakage coexist, low drilling fluid density
8	HD30-H5	Horizontal well	6996	6111.02	Drilling	1.25-1.40	1.37	1.70	Oil and Gas	Throttling circulation	10.67	Oil and gas displacement leading to overflow	Leakage zone in the upper part, limited drilling fluid density
9	MS1	Vertical well	9200	9083.25	Drilling	1.2-2.1	1.91	0.9	Natural gas	Annular pressure back method	5	Drilling into high-pressure oil, gas, and water reservoir	None
10	DQ7	Vertical well	6540	6510	Tripping the drill pipe	1.80-2.00	2.08	0.5	Natural gas	Engineer's method	11	High suction pressure caused by scraper	None
11	YT1	Vertical well	5300	2674	Drilling	1.40-1.70	2.05	0.5	Oil and Gas	Engineer's method, Managed Pressure Drilling	6	High-pressure oil and gas reservoir	None
12	MS504-H12	Horizontal well	8458	7566.98	Drilling	1.10-1.30	1.5	0.30	Oil and Gas	Engineer's method	23.30	Abnormally high-pressure oil, gas, and water reservoir	Inaccurate geological prediction
13	YX3	Horizontal well	8488	8161.77	Drilling	1.1-1.45	1.9	1.1	Oil and Gas	Engineer's method, Managed Pressure Drilling	5.5	Abnormally high-pressure oil, gas, and water reservoir	Inaccurate geological prediction
14	KS8-H1	Directional well	8088	7228.83	Dealing with lost circulation	1.02-1.60	1.05	0.3	Natural gas	Engineer's method	9.5	Treatment of well leakage and density reduction	Prolonged quiescent time, continuous gas intrusion and contamination in the wellbore
15	YG2	Vertical well	8150	6337	Tripping the drill pipe	1.25-1.75	2.5	0.6	Formation brine	Engineer's method	17.4	High-pressure brine intrusion	Poor casing cementing quality
16	MS301-H1	Directional well	8143	8005.2	Drilling	1.12-1.30	1.8	0.6	Natural gas	Engineer's method	21.8	Abnormally high-pressure oil, gas, and water reservoir	Inaccurate geological prediction
17	MS303	Directional well	8084.7	7533.48	Drilling	1.15-1.45	1.75	1.5	Oil and Gas	Engineer's method	26.64	Abnormally high-pressure formation	Inaccurate geological prediction

Conclusions

The integrated intelligent identification technology for "five special" risks (special surface, special lithological bodies, special geological phenomena, special fluids, and special temperature-pressure fields) has been developed. The "five-pressure" (pore pressure, collapse pressure, loss pressure, closure pressure, and fracture pressure) comprehensive prediction technology is adopted to intelligently predict formation pressure.

Five major well control equipment have been developed, including the 175MPa ultra-high pressure blowout preventer stack, 175MPa electric oil nozzle, 175MPa Christmas tree, 175MPa casing head, and 175MPa choke and kill manifold, providing safety guarantees for 10000m ultra-deep wells.

A series of high-performance well control equipment has also been developed, such as high-performance internal blowout prevention tools, liquid-gas separators with a large processing capacity of 1 million cubic meters per day, prototypes of high-grade fully automatic choke and kill systems, ultra-high pressure, large-diameter, low-torque manual-automatic integrated gate valves, and prototypes of ultra-high pressure, full-bore, fast-response new-type choke valves. These equipment ensure the safe disposal of abnormally high-pressure wells.

Comprehensively implement intelligent well control technology management and production organization model optimization at the drilling and completion site. Based on an "expert guidance + professional support + special solutions" approach, the on-site well control problem-solving process is streamlined, well control equipment is containerized, and well control measures are precisely implemented. Digital and intelligent well control equipment and supporting technologies have ensured no well control safety incidents at the SDTK1 well, ensuring safe and efficient exploration and development of oil and gas in the deep carbonate formations of the Tarim Basin.

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References

1. YAN Kai, LI Feng. Research on the well control techniques in Tarim Oilfield[J]. *Progress in Geophysics*, 2008(02):522-527.
2. Wu Xianzhu, Hu Xuguang, Han Liexiang, et al Research progress and prospect of well control technology[J]. *Natural Gas Industry*, 2022, **42**(02):133-142.
3. WU Xianzhu, WAN Fulei, CHEN Zuo, et al Drilling and completion technologies for deep carbonate rocks in the Sichuan Basin: Practices and prospects[J]. *Natural Gas Industry*, 2020, **40**(02):97-105.
4. SONG Zhoucheng; ZHAI Wenbao; DENG Changsong, et al Optimization Technology and Application of Casing Program of Ultra-deep Wells in Fuman Oilfield[J]. *Drilling & Production Technology*, 2022, **45**(06):36-41.
5. Li Qiongyao, Pan Xiaoqing. Prediction Methods of Pore Pressure in Carbonate Rocks[J]. *Shandong Chemical Industry*, 2019, **48**(24):117-118.
6. DUAN Yongxian, SONG Jinpeng, HUAN Zhipeng, YANG Liangang, ZHOU Peng, LV Duanchuan, TIAN Zhihong. Formation, Preservation and Distribution of Abnormally High Pressure in Ordovician Carbonate Rocks in Northern and Central Tarim Basin[J]. *Xinjiang Petroleum Geology*, 2023, **44**(4): 421–428.
7. MA Hai. Pore pressure prediction with seismic interval velocity by the modified Fillippone method[J]. *Petroleum Drilling Techniques*, 2012, **40**(6):56-61.

8. LI Mao, HE Juncai, LI Wei. Comprehensive prediction of formation pressure by multiwell correlation[J]. *Special Oil&Gas Reservoirs*, 2008, **15**(1):88-91.
9. XIONG Jianhua, SHI Yuqin, GUO Min. Effect of tectonic compression on the abnormal pressure of the Feixianguan Formation in the Northeast Sichuan[J]. *Journal of Mineralogy and Petrology*, 2010, **30**(3):83-88.
10. SHI Wanzhong, HE Sheng, CHEN Honghan. Application of joint inverse modeling of multi-seismic-attributes in predicting formation pressure[J]. *Geophysical Prospecting for Petroleum*, 2006, **45**(6):580-585.
11. YU Fu, JIN Yan, CHEN Mian, et al Discussion on a formation pore pressure detection method for carbonate rocks based on the thin plate theory[J]. *Petroleum Drilling Techniques*, 2014, **42**(5):57-61.
12. LU Baoping, BAO Hongzhi, YU Fu. A pore pressure calculating method for carbonate formations based on fluid velocity[J]. *Petroleum Drilling Techniques*, 2017, **45**(3):1-7.
13. MAO Liangjie, CAI Mingjie, WEN Xiaoyong, et al Dynamic simulation of overflow in H2S-containing natural gas well[J]. *Science Technology and Engineering*, 2020, **20**(25):10219-10224.
14. XU Yuqiang, GUAN Zhichuan, PANG Hua, et al A calculation method of the maximum allowable gas invasion overflow in deepwater drilling[J]. *Natural Gas Industry*, 2016, **36**(7):74-80.
15. WU Zhijun, CHEN Gang, LANG Shumin, et al The development of the well control technique in natural gas drilling[J]. *Oil Drilling&Production Technology*, 2010, **32**(5):56-60.
16. LI Yufei, ZHANG Bo, SUN Weifeng. Research on intelligent early kick identification method based on SVM and D-S evidence theory[J]. *Drilling&Production Technology*, 2020, **43**(5):27-30.
17. FAN Xiangyu, Shuai Juntian, Li Zhilin, et al Research status and prospect on early kick detection technology of oil and gas wells[J]. *Drilling&Production Technology*, 2020, **43**(3):23-26.
18. HE Miao, LIU Gonghui, LI Jun, et al A real-time interpretation model for downhole pressure of dual measuring points in managed pressure drilling[J]. *China Petroleum Machinery*, 2019, **47**(10):1-6.
19. ZHANG Bo, HU Xuguang, LIU Guiyi, et al Simulation modeling method of drilling engineering parameters for early overflow monitoring[J]. *Mud Logging Engineering*, 2019, **30**(4):44-50.
20. ZHENG Chongtao, LIU Dianchen, HAN Liexiang. Constant pressure gradient MPD for horizontal wells[J]. *Drilling&Production Technology*, 2019, **42**(3):5-8.
21. TENG Xueqing, SUN Baojiang, ZHANG Yaoming, et al A fivestep bullheading killing well control method for fractured formations without a safety pressure window[J]. *Petroleum Drilling Techniques*, 2018, **46**(6):20-25.
22. WANG Jiangshuai, LI Jun, REN Meipeng, et al A new method for identifying the location of loss zones under managed pressure drilling[J]. *China Petroleum Machinery*, 2020, **48**(9):15-19.
23. ZHANG Guilin. Discussion on the engineer's well killing method in typical blowout cases[J]. *Petroleum Drilling Techniques*, 2018, **46**(6):33-38.
24. ZHANG Ruiyao, LI Jun, LIU Gonghui, et al Automatic throttle control system based on AMESim[J]. *China Petroleum Machinery*, 2021, **49**(4):25-43.
25. LI Qian, YANG Xianlun, WU Xianzhu. A method of controlling the bottomhole pressures in well killing[J]. *Natural Gas Industry*, 2013, **33**(9):87-90.
26. LI Yunhui, HUANG Chuan, CUI Jin, et al Successful application of killing well tech using replacement method in Heba 1 Well in Northeast of Sichuan[J]. *Well Testing*, 2008, **17**(4):45-46.
27. ZHANG Xingquan, LI Xiangfang, REN Meipeng, et al Research on the well control techniques of bullheading[J]. *Petroleum Drilling Techniques*, 2012, **40**(3):62-66.

28. GONG Peibin, SUN Baojiang, WANG Zhiyuan, et al A study on well killing method of empty blowout wells[J]. *Journal of Oil and Gas Technology*, 2012, **34**(1):100-103.